

Solving Mystery Sequences with Polynomials

Workshop in support of Girls' Math Day 2025, MathCounts Outreach

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Girls' Math Day 2025

Introduction:

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This sequence appears in nature: the golden spiral.

1, 1, 2, 3, 5, ...?

These problems are incredibly **unfair**: after all, there is nothing stopping any number you can think of from coming next. It's ambiguous!

Mystery Polynomials

To help remedy this, we will assume that our sequences are **generated** by some mystery **polynomial** $f(x) = ax^n + bx^{n-1} + \dots$ that we don't know.

The 1st entry of the sequence will be $f(1)$, the 2nd entry will be $f(2)$, and so on.

For example, if $f(x) = x^2 + 1$ then our sequence would be:

$$f(1), f(2), f(3), \dots = 1^2 + 1, 2^2 + 1, 3^2 + 1, \dots = 2, 5, 10, \dots$$

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So, using only the *outputs* (y -values) of f at $x = 1, 2, 3, \dots$, is there a way to find $f(x)$ or find some future value $f(k)$ using math instead of guessing?

The mathematical name for this is **Polynomial Interpolation** and it's an **extremely** important idea in Biology, Statistics, Machine Learning, and Computer Science.

Using Algebra

Sometimes you can easily guess the next term:

$$2, 5, 8, \dots?$$

But we want to use **Algebra** to make this guess rigorous.

Give me a polynomial that creates the sequence: $2, 5, 8, \dots$. It must satisfy $f(1) = 2, f(2) = 5, f(3) = 8, \dots$

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$$2 = m \times 1 + b$$

$$5 = m \times 2 + b$$

- Subtract both equations to get $m = 3$, substitute for $b = 1$. Why does this make sense **intuitively**?

Pause & Ponder

Here are some more difficult problems: work together and see if you can find the mystery polynomials!

Quadratic Polynomial

Alice knows the function $P(x) = x^2 + ax + b$, but she doesn't tell us what a and b are. However, she does tell us that $P(1) = 2$ and $P(2) = 7$. What is $P(3)$?

Hint: Set up a system of linear equations for a and b , then solve for them.

Cryptography

Charlie's phone password is a four-digit number. He writes the first two digits as m and the second two digits as b and makes the function $f(x) = mx + b$. What value of x should you ask Charlie for $f(x)$ to find his password?

Hint: $1234 = 1200 + 34$. Never store your passwords as linear functions.

Danger!

When solving hard problems, it's very good to **try some small cases**, since they might be easier, and try to guess a pattern for larger/harder cases.

For example if you notice that:

$$1 = 1^2, 1 + 3 = 2^2, 1 + 3 + 5 = 3^2, \dots$$

You could then try to prove more rigorously that $1 + 3 + \dots (2n - 1) = n^2$. Here's an example where it goes astray...

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Moser's Circle Problem

- If you draw 2 points on a circle and connect them, how many regions do you get?
- What about connecting 3 points on the circle with lines? How many regions?
- What's the maximum number of regions you can get with 4 points?

Can anyone spot the pattern?

Danger!

Finding the number of regions with n points is not easy, but here is the sequence:

$$1, 2, 4, 8, 16, \mathbf{31}, \dots$$

What!? It's not powers of 2? It turns out that this sequence, even though you might think you know the pattern, is actually more complex.

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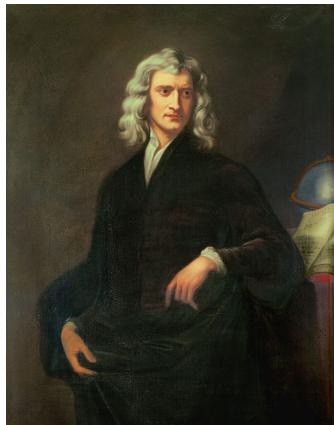
Here's the mystery polynomial that generates this sequence (this can be **proven** combinatorially, but it is tricky):

$$P(x) = \frac{1}{24}(24 - 18x + 23x^2 - 6x^3 + x^4)$$

$$\text{For example, } P(2) = \frac{24 - 18 \times 2 + 23 \times 4 - 6 \times 8 + 16}{24} = \frac{48}{24} = 2.$$

Newton's Method of Divided Differences

You probably know Sir Isaac Newton:



A brilliant physicist and mathematician who came up with many extremely important ideas: orbital motion, **Calculus!**, gravity, optics (study of light), and more.

Did you know that Newton tried to solve mystery sequences too? Let's look at his method (**Gregory-Newton Interpolation Formula**).

Newton's Method of Repeated Differences

Let's showcase this beautiful technique! Start with a mystery sequence:

$$5, 9, 15, 23, \dots$$

Then, find out how much to add to get the next term:

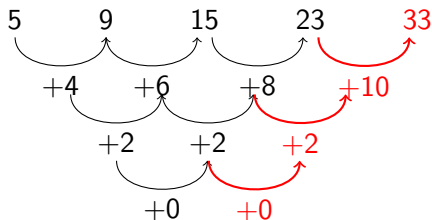
$$\begin{array}{ccccccc} 5 & & 9 & & 15 & & 23 \dots \\ & \swarrow & & \swarrow & & \swarrow & \\ & +4 & & +6 & & +8 & \end{array}$$

Keep doing this until you finish your (upside down) pyramid: it should end at **0**.

$$\begin{array}{ccccccc} 5 & & 9 & & 15 & & 23 \\ & \swarrow & & \swarrow & & \swarrow & \\ & +4 & & +6 & & +8 & \\ & & \swarrow & & \swarrow & & \\ & & +2 & & +2 & & \\ & & & \swarrow & & \swarrow & \\ & & & +0 & & & \end{array}$$

Method of Repeated Differences

As we go down the pyramid, it becomes easier and easier to spot the pattern. To find the next entry, we just extend the arrows working from **bottom to top**:



Thus, for any mystery sequence generated by a polynomial, we have an **algorithmic** way to continue it indefinitely.

Using Algebra (Again)

We can also use algebra to solve this problem. Instead of using $f(x) = mx + b$ as our template, we will use $f(x) = ax^2 + bx + c$. We still don't know a, b, c so we try to solve the system:

$$f(1) = 5 \implies a + b + c = 5$$

$$3a + b = 9 - 5$$

$$f(2) = 9 \implies 2^2a + 2b + c = 9$$

$$5a + b = 15 - 9$$

$$f(3) = 15 \implies 3^2a + 3b + c = 15$$

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After more subtracting of the equations we get $a = 1, b = 1, c = 3$, so $f(x) = x^2 + x + 3$ and let's check that $f(4) = 4^2 + 4 + 3 = 23$ (that's the 4th term) and $f(5) = 5^2 + 5 + 3 = 33$ (which matches our pyramid). Thus $f(x) = x^2 + x + 3$ is the mystery polynomial!

Even though we had to “guess” that f was a quadratic, we used our guess to then make a more rigorous calculation. This is called an “**ansatz**”: a guess about the solution that is then later verified.

Historical Conjecture...

- Notice that the second-to-last layer is a **constant** function ($y = c$).
- The layer above it grows equally at each step: it's a **linear** function ($y = mx + b$).
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When you study Calculus in the coming years, you will learn about **differentiation**, ie $\frac{d}{dx}$, which turns **quadratics into linear functions into constant functions** (makes polynomials simpler).

In fact, here is the definition of a derivative from a Calculus textbook:
 $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$. (*Don't worry if this doesn't make sense*) And here is our formula for a difference in our pyramid: $\frac{f(x+1) - f(x)}{1}$.

Don't they look similar? Do you think it's interesting that Newton invented Calculus *and* the Difference Method of "mystery sequence solving"?

Repeated Difference Practice

Now, it's your turn! Work together and use the method of repeated differences to...

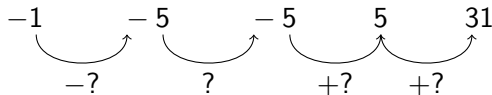
Sequence Manipulation

Find the next element of the sequence:

$$-1, -5, -5, 5, 31, \dots?$$

Bonus Problem: Can you find the element that came **before** the -1 ?

It should start something like this:



And continue until you hit 0...

More Advanced Method

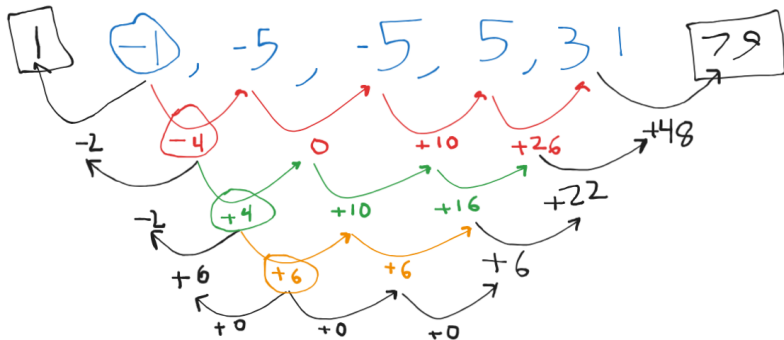


Figure: Does your paper look like this?

Using the difference pyramid allows us to find past/future sequences, but how do we find the formula for the polynomial itself?

Use the circled numbers at the left: $-1, -4, +4, +6$.

More Advanced Method

Newton's formula uses those circled numbers in a very interesting way:

$$P(n) = -1 + -4 \frac{(n-1)}{1} + +4 \frac{(n-1)(n-2)}{1 \times 2} + +6 \frac{(n-1)(n-2)(n-3)}{1 \times 2 \times 3}$$

This looks pretty complicated! But do you see the pattern?

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- Let's try plugging in $n = 1$:

$$-1 - 4 \frac{(1-1)}{1} + \frac{(1-1)(1-2)}{1 \times 2} + 6 \frac{(1-1)(1-2)(1-3)}{1 \times 2 \times 3} = -1. \text{ Notice how everything cancels except for the first term?}$$

- Same for $n = 2$:

$$-1 - 4 \frac{(2-1)}{1} + \frac{(2-1)(2-2)}{1 \times 2} + 6 \frac{(2-1)(2-2)(2-3)}{1 \times 2 \times 3} = -1 - 4 = -5. \text{ This time everything but the first and second terms canceled.}$$

So, without doing any algebra, we can still find out the mystery polynomial! (You still have to expand out the parentheses and collect terms though...)

Challenge Problems

If time permits, feel free to talk to me about these problems or try them on your own!

Challenge 1:

Alice picks a new secret polynomial $f(x) = x^2 + ax + b$ and tells us that $f(1) = 9$. What's the largest possible value for $a \times b$?

Hint: Think of a rectangle of side lengths a, b . What is its area & perimeter?

Challenge 2:

If our mystery polynomial is $f(x) = ax^n + bx^{n-1} + \dots$ (ie a degree n polynomial), how many terms of the sequence $f(1), f(2), f(3) \dots$ will you need in order to solve for all of the coefficients $a, b, c, \dots$?